MATH730: Algebraic Topology

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1. Motivation and Review

Motivation **1.1**

Distinguish topological objects, and ultimately classify them.

Let's see some examples where the power of algebraic topology helps us prove important topological properties.

Theorem **1.2**

The unit ball $\overline{\mathbb{D}}^{n+1}$ is not homeomorphic to S^n

Proof. The nth homology group H_n distinguishes them. Namely, $H_n\left(\overline{\mathbb{D}}^{n+1}\right) = \{0\}$, while $H_n(S^n) = \mathbb{Z}$. □

Theorem **1.3**

There does not exist a continuous map $f : \overline{\mathbb{D}}^{n+1} \to S^n$ such that $f(x) = x$ for every $x \in S^n$ (note that S^n is the boundary of $\overline{\mathbb{D}}^{n+1}$).

Proof. Suppose such an f exists. Consider the inclusion map $i : S^n \to \overline{\mathbb{D}}^{n+1}$. Then, the map $f \circ i : S^n \to S^n$ is the identity map. Moreover, i and f induce maps on the homology groups

$$H_n(S^n) \xrightarrow{H_n(i)} H_n(\mathbb{D}^{n+1}) \xrightarrow{H_n(f)} H_n(S^n).$$

By functoriality,

$$H_n(f) \circ H_n(i) = H_n(f \circ i) = H_n(\mathrm{id}) = \mathrm{id}\,.$$

Hence, we have

$$\mathbb{Z} \to 0 \to \mathbb{Z}$$

induces the identity map $\mathbb{Z} \to \mathbb{Z}$, which is impossible. □

Brauer's Fixed Point Theorem **1.4**

Every continuous function $f : \overline{\mathbb{D}}^n \to \overline{\mathbb{D}}^n$ has a point x such that $f(x) = x$.

Proof. Suppose f does not have a fixed point. Then, for every $x \in \overline{\mathbb{D}}^n$, we may consider the unique straight line from $f(x)$ to x. Let $g(x)$ be the point where the line $f(x) \to x$ hits S^n. Observe that $g(x) = x$ for every $x \in S^n$, and g is continuous. Thus, we have a contradiction by Theorem 1.3. □

1.1. Topology Review

Definition 1.5

In $\mathbb{R}^n$, we have the usual metric $d(x,y)$. We denote $B(x,r)$ to be the open ball around x with radius r.

Definition 1.6

A topological space X is called **compact** if every open cover has a finite subcover.

Theorem 1.7

A subset $X \subset \mathbb{R}^n$ is compact if and only if X is closed and bounded.

Proof. Suppose X is compact. The set $\{B(0,n)\}_{n=1}^{\infty}$ of open balls is necessarily an open cover of X, hence has a finite subcover, thus X is bounded. Moreover, if $\mathbb{R}^n - X$ is not open, let $p \in \mathbb{R}^n \setminus X$ be such that $B(p,\varepsilon) \cap X \neq \emptyset$. Then, take the cover of X by

$$\left\{B(x, \frac{d(p,x)}{2})\right\}_{x \in X}.$$

One sees that this does not have a finite subcover, a contradiction.

For the other direction, note that closed+bounded implies sequential compactness. Let $\{U_\alpha\}_{\alpha \in I}$ be an open cover of X.

Claim 1.8. *There exists $\delta > 0$ such that for every $x \in X$, $B(x,\delta) \subset U_\alpha$ for some $\alpha \in I$.*

Proof. Contrarily, if this were not the case, there exists a sequence $\{x_i\} \subset X$ such that $B(x,1/n) \not\subset U_\alpha$ for all $\alpha \in I$. Then, $\{x_i\}$ has a subsequence converging to some p. This p must be contained in some U in the open cover, and it follows that there exists a δ and N with $B(x_N,\delta) \subset U$. □

Since X is bounded in $\mathbb{R}^n$, for every $\delta > 0$ the number of disjoint balls of radius δ around points in X is bounded. Let m be this upper bound. Take a point $p_1 \in X$. The ball $B(p,\delta)$ is contained in some cover element U_1. If U_1 covers X, we're done, but otherwise repeat with some p_2 and U_2. Continue to repeat until we have p_{m+1} and U_{m+1}. We claim that $U_1, \ldots, U_{m+1}$ must cover X: indeed, if it doesn't, then the balls $\{B(p_i,\delta/2)\}_{i=1}^{m+1}$ must all be disjoint, and this is a contradiciton.

□

Definition 1.9

A topological space X is called **connected** if $X = U_1 \sqcup U_2$ with U_1, U_2 open implies $U_1 = \emptyset$ or $U_2 = \emptyset$.

Definition 1.10

X is called **path connected** if for every $p,q \in U$, there exists a continuous map $\delta : [0,1] \to X$ such that $\delta(0) = p$ and $\delta(1) = q$.

Proposition 1.11

Path connected $\implies$ connected.

Definition 1.12

A topological space X is called **Hausdorff** if for every distinct $p, q \in X$, there exists $p \in U$ and $q \in V$ open such that $U \cap V = \emptyset$.

Theorem 1.13

Let $f : X \to Y$ be bijective and continuous, with X compact, Y Hausdorff. Then, f is a homeomorphism.

Proof. It suffices to show f is an closed map. Let $C \subset X$ be closed. Since X is compact, one can see that C is compact (theorem). Therefore, $f(C)$ is compact (theorem), hence closed (theorem). □

Proposition 1.14

If X is Hausdorff and $A \subset X$ is compact, then A is closed.

Exercise 1. *Give an example of a space X such that $A \subset X$ is compact but not closed. This space cannot be Hausdorff.*

Definition 1.15

Let X be a topological space, and $\sim$ be an equivalence relation on X. The **quotient-topology** is defined as follows: let $\pi : X \to X/\sim$ be the projection map. Then, $U \subset X/\sim$ is open whenever $\pi^{-1}(U)$ is open.

Proposition 1.16

A map $f : (X/\sim) \to Y$ is continuous if and only if $f \circ \pi$ is continuous.

Example 1.17. *Let $\mathbb{RP}^n$ be the space of lines through the origin in $\mathbb{R}^{n+1}$: this is the quotient space of $\mathbb{R}^{n+1} \setminus \{0\}$ by the equivalence $x \sim y \iff$ there exists $\lambda \in \mathbb{R}$ with $\lambda x = y$. This is homeomorphic to the quotient space of S^n by the relation $x \sim -x$.*

Definition 1.18

Let $(X_\alpha, \mathcal{U}_\alpha)_{\alpha \in I}$ be topological spaces. The **product topology** on $\prod_I X_\alpha$ has a basis of sets of the form

$$\prod_{\alpha \in U} U_\alpha, \quad U_\alpha \in \mathcal{U}_\alpha,$$

and $U_\alpha = X_\alpha$ for all but finitely many $\alpha \in I$.

***Tychonoff's Theorem* 1.19**

The product of compact spaces is compact.

2. CW Complexes

Definition 2.1

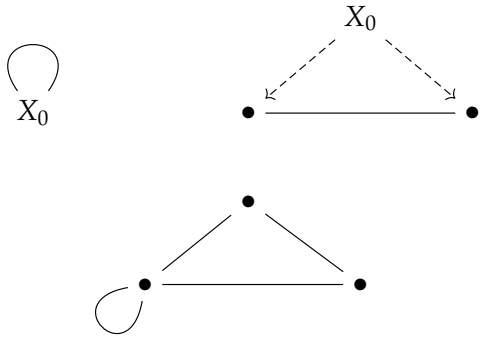
A **CW complex** is a topological space X, together with a sequence of subspaces $X_0 \subset X_1 \subset \cdots$ such that

1. $X = \bigcup X_n$
2. $U \subset X$ is open if and only if $U \cap X_n$ is open for all $n \geq 0$
3. X_0 is a discrete set of points
4. For every $n \geq 1$, there exists an indexing set $I(n)$, disks $\{D_\alpha^n\}_{\alpha \in I(n)}$, and continuous maps $\varphi_\alpha^n : \partial D_\alpha^n \to X_{n-1}$ such that
$$X_n = \bigsqcup_{\alpha \in I(n)} \overline{D}_\alpha^n \sqcup X_n / \sim$$
where $x \sim \varphi_\alpha^n(x)$ for all $\alpha \in I(n)$, $x \in \partial D_\alpha^n$.

We call X_n the nth **skeleton** of X, and φ_α^n is called an **attaching map**. For a CW complex X, the sets $\{X_n\}$ and $\{\varphi_\alpha^n\}$ are called a **CW structure** on X. The quotient maps $\Phi_\alpha^n : \overline{D}_\alpha^n \to X_n$ are called the **characteristic maps**. If $X_{n-1} \subsetneq X_n$, but $X_i = X_j$ for all $i, j \geq n$, we say that X is **n-dimensional**.

Example 2.2. *A discrete set of points is a* 0*-dimensional CW complex.*

Example 2.3. *A graph is a* 1*-dimensional CW complex.*



Example 2.4. *$\mathbb{R}^n$ is a CW complex. For example, $\mathbb{R}^2$ has the* 0*-cell $\mathbb{Z}^2$, the* 1*-cell a $\mathbb{Z}^2$-grid, and the* 2*-cell induced by attaching each disk to the square of the grid.*

Example 2.5. *A torus is a CW complex.*

Example 2.6. *S^n is a CW complex. Namely, X_0 is a single point, $X_1 = \cdots = X_{n-1} = X_0$, and the (only possible) attaching map $\varphi_1^n : D_1^n \to X_{n-1}$.*

Example 2.7. *$\mathbb{RP}^n = S^n/(x \sim -x)$ is a CW complex. When $n = 0$, $\mathbb{RP}^n$ is just a point, hence as a CW structure. Suppose $\mathbb{RP}^{n-1}$ has a CW structure. Let $\overline{D}_\alpha^n$ be the upper hemisphere. We obtain the CW structure on $\mathbb{RP}^n$ by attaching one n-cell to $\mathbb{RP}^{n-1}$ via the attaching map*
$$\varphi_\alpha^n(x_1, \ldots, x_n, 0) = [(x_1, \ldots, x_n)] \subset \mathbb{RP}^{n-1}.$$

Exercise 2. *Find a CW structure on* $\mathbb{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\})/(x \sim \lambda x, \lambda \in \mathbb{C}^*)$

Lemma 2.8

Let X be a CW-complex. Then, $U \subset X$ is open $\iff$ for every n and $\alpha \in I(n)$, $(\Phi_\alpha^n)^{-1}(U)$ is open.

Proof. Recall that U is open in X if and only if $U \cap X^n$ is open for all n, if and only if for the projection map

$$P^n : X^n \sqcup \bigsqcup_{\alpha \in I(n)} D_n^\alpha \to X^n,$$

we have

$$(P^n)^{-1}(U \cap X^n)$$

is open. This is the case if and only if for each $\alpha \in I(n)$,

$$(\Phi_\alpha^n)^{-1}(U \cap X)$$

and $U \cap X^{n-1}$ are open. So, ($\implies$) is verified, and ($\impliedby$) follows by induction on n. □

Lemma 2.9

Let $K \subset X$ be compact. Then, K intersects only finitely many cells in X.

Proof. Suppose not. Then, we can choose an infinite set $S \subset K$ such that no two points in S belong to the same cell. We claim S is closed. Indeed, note that $S \cap X^0$ is closed since X^0 is discrete. By induction, suppose $S \cap X^{n-1}$ is closed. Then, $S \cap X^n$ is closed $\iff$ $(\Phi_\alpha^n)^{-1}(S)$ is closed, and by induction $(\Phi_\alpha^n)^{-1}(S) \cap \partial D_\alpha^n$ is closed, so indeed $(\Phi_\alpha^n)^{-1}(S)$ is closed as the union of closed sets. Thus, $S \cap X^n$ is closed for every n, hence S is closed.

In fact, we've shown that every subset of S is closed, so S is discrete. Moreover, S is compact as a closed subset of a compact set. Thus, S is compact and discrete, hence finite. □

Definition 2.10

Let $A \subset X$ be a set, and fix $\varepsilon := \{\varepsilon_\alpha^n \in \mathbb{R}^+\}$ for every n and $\alpha \in I(n)$. Then, an **ε-neighborhood** of A is the open set $N_\varepsilon(A) \subset X$ defined as follows: let

$$N_\varepsilon^0(A) := A \cap X^0$$

and then for every $\alpha \in I(n)$, set

$$(\Phi_\alpha^n)^{-1}(N_\varepsilon^n(A)) := [B(A \cap \mathrm{Int}(D_\alpha^n), \varepsilon_\alpha^n) \cap \mathrm{Int}(D_\alpha^n)] \cup \left[(1 - \varepsilon_\alpha^n, 1](\Phi_\alpha^n)^{-1}\left(N_\varepsilon^{n-1}(A)\right)\right],$$

and

$$N_\varepsilon(A) = \bigcup N_\varepsilon^n(A).$$

Proposition 2.11

$$N_\varepsilon(A) \cap X^n = N_\varepsilon^n(A)$$

Theorem 2.12

Let X be a CW complex. Then, X is Hausdorff.

Proof. Let $p \neq q \in X$, and suppose that $p, q \in X^n$ but $q \notin X^{n-1}$. So, q is in some disk e^n_α. If $p \in X^{n-1}$, we may choose ε sufficiently small so that $N_\varepsilon(X^{n-1})$ and $N_\varepsilon(q)$ are disjoint. If $p \notin X^{n-1}$, then p, q are distinct points in e^n_α which is Hausdorff, hence separable in ε-neighborhoods. These ε-neighborhoods lift to the global structure of X so that $N_\varepsilon(p) \cap N_\varepsilon(q) = \emptyset$. □

Definition 2.13

Let X, Y be topological spaces. Two (continuous) maps $f, g : X \to Y$ are **homotopic** if there exists $H : X \times [0,1] \to Y$ continuous such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for every $x \in X$.

For a subset $A \subset X$, two maps $f, g : X \to Y$ are called **homotopic relative to A** if $f|_A = g|_A$, and there exists a homotopy $H : X \times [0,1] \to Y$ between f, g such that $H(x,t) = f(x) = g(x)$ for all $x \in A$, $t \in [0,1]$.

Homotopy is denoted $f \sim g$ or $f \overset{H}{\sim} g$.

Definition 2.14

Two maps $f : X \to Y$ and $g : Y \to X$ are called **homotopy equivalent** if $g \circ f \sim \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. In this case, X and Y are said to be **homotopically equivalent.**

A subset $A \subset X$ is called a **retract** of X if there exists $r : X \to A$ continuous such that $r(x) = x$ for every $x \in A$. Moreover, A is called a **deformation retract** if $r \sim \mathrm{id}_X$ relative to A. X is called **contractible** if $\mathrm{id}_X \sim$ constant

Example 2.15. *Show that $\mathbb{R}^{n+1} \setminus \{0\}$ deformation retracts to $S^n \subset \mathbb{R}^{n+1}$.*

Example 2.16. *Show that $i : \mathbb{R}^n \to \mathbb{R}^{n+k}$ and $\pi : \mathbb{R}^{n+k} \to \mathbb{R}^n$ are homotopy equivalent.*

Example 2.17. *Let X be a topological space. Define $C(X) = X \times [0,1]/[(x,1) \sim (y,1)]$. Show that $C(X)$ is contractible.*

Definition 2.18

X is called **locally contractible** if for every $p \in X$ and $U \ni p$ open, there exists $V \subset U$ with $p \in V$ such that p is contractible.

Proposition 2.19

Every CW complex is locally contractible.

Proof. Exercise. Inductively push the additional part of the neighborhood to the boundary. □

Corollary 2.20

If X is a CW complex, then X is connected $\iff$ it's path connected.

Proof. Exercise. Local contractibility allows you to understand path connectivity because of the path induced by a retract. □

3. Fundamental Group

Definition 3.1

Let X be a topological space $p \in X$. The **fundamental group** of X at p, is

$$\pi_1(X,p) := \{\gamma : [0,1] \to X \mid \gamma(0) = \gamma(1)\} / \left[\gamma \overset{\text{rel}(\{0,1\})}{\sim} \mu\right].$$

Equivalently, let q be a fixed point in S^1, and we have

$$\pi_1(X,p) = \left\{\gamma : S^1 \to X\right\} / \left[\gamma \overset{\text{rel}(q)}{\sim} \mu\right]$$

This definition is in some ways better, since changing S^1 to S^n gives a more generalized definition of the ***n*-th fundamental group** $\pi_n(X,p)$.

Definition 3.2

A connected space X is called **simply connected** if there exists $p \in X$ such that $\pi_1(X,p) = \{[p]\}$.

Proposition 3.3

If X is contractible, then

$$\pi_1(X,p) = \{[p]\}$$

for the p such that X deformation retracts to p.

Proof. Let $\gamma \in \pi_1(X,p)$. Let H be a homotopy between id_X and p. Take

$$\widetilde{H}(s,t) := H(\gamma(s),t).$$

Then, $\gamma \overset{\widetilde{H}}{\sim} p$. □

Definition 3.4

For $\gamma, \mu \in \pi_1(X,p)$, define

$$\gamma \star \mu(t) = \begin{cases} \mu(2t) & 0 \le t \le 1/2 \\ \gamma(2t-1) & 1/2 \le t \le 1 \end{cases}.$$

Then, $\star$ induces a well-defined group structure on $\pi_1(X,p)$.

Indeed, in order to show that this concatenation of curves defines a group structure on $\pi_1(X,p)$, one must prove the following claim:

Claim 3.5. *If $[\gamma] = [\gamma']$ and $[\mu] = [\mu']$, then $[\gamma \star \mu] = [\gamma' \star \mu']$. Thus, $[\gamma] \star [\mu]$ is a well-defined binary operation on $\pi_1(X,p)$. Moreover, $\pi_1(X,p)$ is a group, with identity the constant map $[p]$, and*

$[\gamma(t)]^{-1} = [\gamma(1-t)]$.

Exercise 3. *Show that if $\varphi : [0,1] \to [0,1]$ is a continuous map with $\varphi(0) = 0$ and $\varphi(1) = 1$, then $\gamma \circ \varphi \sim \gamma$.*

Theorem 3.6

Let X be path connected, $p, q \in X$. Then, $\pi_1(X,p) \cong \pi_1(X,q)$.

Proof. Since X is path connected, we have a continuous curve $\mu : [0,1] \to X$ with $\mu(0) = p$ and $\mu(1) = q$.

Claim 3.7. *Let Ω_p denote the space of curves starting and ending at p. Then, the map*

$$T_\mu : \Omega_p \to \Omega_q$$

defined by $T_\mu := \mu \star \gamma \star \mu^{-1}$ induces an isomorphism $\pi_1(X,p) \xrightarrow{\sim} \pi_1(X,q)$.

Indeed, suppose $[\gamma] = [\gamma']$. Then, $[T_\mu(\gamma)] = [T_\mu(\gamma')]$. Moreover,

$$\mu \star (\gamma_1 \star \gamma_2) \star \mu^{-1} = \mu \star \gamma_1 \star \mu^{-1} \star \mu \star \gamma_2 \star \mu^{-1},$$

hence T_μ is a homomorphism. Moreover, the map $\psi = \mu(1-t)$ induces a clear inverse homomorphism $T_\psi : \pi_1(X,q) \to \pi_1(X,p)$, so it follows that T_μ is an isomorphism. □

Proposition 3.8

Let (X,p) and (Y,q) be (pointed) topological spaces, and $f : (X,p) \to (Y,q)$ be continuous (i.e. a continuous map $f : X \to Y$ with $f(p) = q$). Then, the map $f_* := \pi_1(f) : \pi_1(X,p) \to \pi_1(Y,q)$ defined by

$$f_*([\gamma]) = [f \circ \gamma]$$

is a well-defined group homomorphism. Moreover, if

$$(X,p) \xrightarrow{f} (Y,q) \xrightarrow{g} (Z,r)$$

is a chain of continuous maps, then $(g \circ f)_* = g_* \circ f_*$.

Proof. Let $[\gamma] = [\gamma']$. Then, $f \circ \gamma \sim f \circ \gamma'$ on $f \circ H$, relative to $\{0,1\}$, so this is well-defined, and

$$(f \circ \gamma) \star (f \circ \mu) = f(\gamma \star \mu)$$

at the curves level, which induces the homomorphism $(g \circ f)_* = g_* \circ f_*$ at the level of fundamental groups. □

***Functoriality of the Fundamental Group* 3.9**

The map π_1 on spaces and the continuous maps between them is a functor from the category of pointed spaces to the category of groups. That is,

$$\pi_1 : \mathrm{Top}_\bullet \to \mathrm{Grp}$$

is a covariant functor.

Definition 3.10

X is called **strongly contractable** if there exists $p \in X$ such that X deformation retracts to p.

Proposition 3.11

If X is strongly contractable, then $\pi_1(X, q) = \{e\}$ for all $q \in X$.

Proof. When $q = p$, we know $\mathrm{id}_X \overset{\mathrm{rel}\, p}{\sim} (x \mapsto p)$, and it follows that $\pi_1(X, p)$ is trivial. Then, the theorem follows for arbitrary q by Theorem 3.6, since strongly contractable implies path connected. □

Recall

X is called contractible if $X \sim \{q\}$, i.e. $\mathrm{id}_X \sim (x \mapsto q)$.

Theorem 3.12

If X is contractible, then $\pi_1(X, q) = \{e\}$ for all $q \in X$.

4. The Fundamental Group of S^1

Theorem 4.1

Let $\gamma : [0,1] \to S^1$ be continuous with $\gamma(0) = 1$. Then, there exists a unique $\theta : [0,1] \to \mathbb{R}$ such that $\theta(0) = 0$, and

$$e^{i\theta(t)} = \gamma(t).$$

Moreover, we can do this continuously along homotopies. That is, if $H : [0,1] \times [0,1] \to S^1$ has $H(0, s) = 1$, there there exists a unique $\theta : [0,1] \times [0,1] \to \mathbb{R}$ such that $\theta(0, s) = 0$ for all s, and $e^{i\theta} = H$ for all t, s.

Corollary 4.2

Let $\gamma_1, \gamma_2 : [0,1] \to S^1$ be such that $\gamma_i(0) = \gamma_i(1) = 1$, $i = 1, 2$. Let θ_1, θ_2 be their lifts as in Theorem 4.1. Then,

$$\gamma_1 \overset{\mathrm{rel}\{0,1\}}{\sim} \gamma_2 \iff \theta_1(1) = \theta_2(1)$$

Proof. ($\implies$) Suppose $\gamma_1 \sim \gamma_2$ relative to $\{0,1\}$, via a homotopy H. Let $\theta : [0,1] \times [0,1] \to \mathbb{R}$ be the lift of H as in Theorem 4.1. Then, $\theta_1(x) = \theta(x, 0)$ and $\theta_2(x) = \theta(x, 1)$. Consider the path $s \mapsto \theta(1, s)$. Then, since $H(1, s) = 1$, and $e^{i\theta} = H$, we have $\theta(1, s) \in 2\pi\mathbb{Z}$. So, as $\theta(1, s)$ is continuous with a discrete image, it must be constant, hence $\theta_1(1) = \theta_2(1)$.

($\impliedby$) Take $\theta(t, s) = (1 - s)\theta_1(t) + s\theta_2(t)$. Then, $\theta(t, s)$ is a homotopy between θ_1 and θ_2, with $\theta(1, s) = \theta_1(1)$, and $\theta(0, s) = 0$. Now, we may take $H(t, s) := e^{i\theta(t,s)}$, and H is the desired homotopy from γ_1 to γ_2 relative to $\{0,1\}$. Note that relative to 1 comes from the fact $\theta_1(1) = \theta_2(1)$. □

Corollary 4.3

For any $p \in S^1$,

$$\pi_1(S^1, p) \cong \mathbb{Z},$$

with $t \mapsto e^{2\pi i t} : [0,1] \to S^1$ being the generator.

Proof. Clearly, $(e^{2\pi i t})^k \overset{\text{rel}\{0,1\}}{\sim} e^{2\pi i k t}$, where the LHS denotes concatenation k times. On the other hand, each element in $\pi_1(S^1, 1)$ is in bijective correspondence with $t \mapsto e^{2\pi i k t}$ for some $k \in \mathbb{Z}$. Indeed, if θ is the lift of an element in $\pi_1(S^1, 1)$, then $\theta_1(1) \in 2\pi\mathbb{Z}$, and the bijective correspondence is induced by $\gamma \leftrightarrow \frac{\theta_1(1)}{2\pi}$. □

Exercise 4. *Let $\{(X_i, p_i)\}_{i=1}^N$ be a set of pointed spaces. Show that*

$$\pi_1\left(\prod_{k=1}^N X_i, (p_1, \ldots, p_n)\right) \cong \prod_{k=1}^N \pi_1(X_i, p_i).$$

Exercise 5. *Show that a continuous map $f : \overline{D}^2 \to S^1$ satisfying $f(x) = x$ for all $x \in S^1$ does not exist.*

Proposition 4.4

For all $k \geq 2$, $\pi_1(S^k, p)$ is trivial.

Proof. Let $\gamma : [0,1] \to S^k$ be continuous. We claim γ is homotopic, relative to $\{0,1\}$, to a curve $\widetilde{\gamma}$ that misses a point q. The proposition follows from the claim: indeed, we have $\psi : S^k \setminus \{q\} \xrightarrow{\sim} \mathbb{D}^k$, so consider $\mu := \pi \circ \widetilde{\gamma}$, which is homotopic to the constant map, and we may push it back to S^k via ψ^{-1} to show $[\gamma] = [\widetilde{\gamma}] = [\text{id}]$.

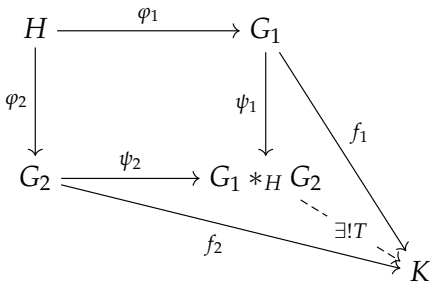
To prove the claim, we approximate γ via smooth maps, and smooth maps always miss points by Sard's theorem. □

5. Amalgamated Products and Free Products

Definition 5.1

Let H, G_1, G_2 be groups, and let $\varphi_i : H \to G_i$ be homomorphisms. The **amalgamated product** of G_1 and G_2 along H, denoted $G_1 *_H G_2$, is a group together with homomorphisms $\psi_i : G_i \to G_1 *_H G_2$ satisfying $\psi_1 \circ \varphi_1 = \psi_2 \circ \varphi_2$. Moreover, it has the universal property that for every group K and homomorphisms $f_i : G_i \to K$ with $f_1 \circ \varphi_1 = f_2 \circ \varphi_2$, there exists a unique homomorphism $T : G_1 *_H G_2 \to k$ such that $T \circ \psi_i = f_i$. That is, the following diagram

commutes:



A good way to think about this is gluing G_1 and G_2 along H.

Example 5.2. *Consider a similar definition but with sets $A \xrightarrow{\varphi_1} B$ and $A \xrightarrow{\varphi_2} B_2$. If we have $f_1 : B_1 \to K$ and $f_2 : B_2 \to K$, then one can see $B_1 \sqcup B_2$ with the natural embeddings $i_1 : B_1 \to B_1 \sqcup B_2$, $i_2 : B_2 \to B_1 \sqcup B_2$, has the desired unique $\widetilde{T} : B_1 \sqcup B_2 \to K$. However, we do not have the desired commutativity $i_1 \circ \varphi_1 = i_2 \circ \varphi_2$. In order to have this, we need to quotient $B_1 \sqcup B_2$ by the relation $i_1\varphi_1(x) \sim i_2\varphi_2(x)$. Then, $(B_1 \sqcup B_2)/\sim$ is the amalgamated product of these sets.*

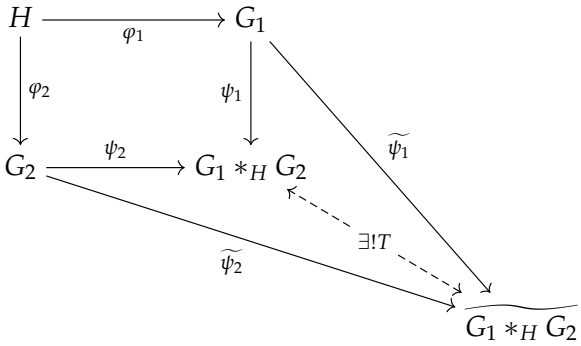
Existence and Uniqueness of the Amalgamated Product 5.3

Let H, G_1, G_2 be groups with homomorphisms $\varphi_i : H \to G_i$. Then, the amalgamated product $(G_1 *_H G_2, \psi_1, \psi_2)$ exists, and if $(\widetilde{G_1 *_H G_2}, \widetilde{\psi}_1, \widetilde{\psi}_2)$ is another amalgamated product, then there exists a unique isomorphism

$$T : G_1 *_H G_2 \to \widetilde{G_1 *_H G_2}$$

such that $T \circ \psi_i = \widetilde{\psi}_i$.

Proof.



□

Definition 5.4

Let G_1, G_2 be two groups. The **free product** $G_1 * G_2$ is the group of words in $G_1 \setminus \{e_1\}, G_2 \setminus \{e_2\}$, and a new identity $\{e\}$, where elements from the same group multiply as expected, the product of elements is concatenation, and multiplying an element g from G_1 or G_2 by its inverse in that group yields e.

Theorem 5.5

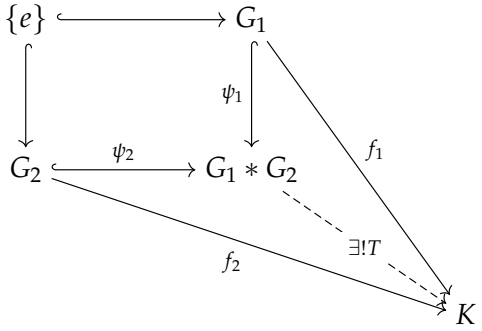
Let G_1, G_2 be groups. Then,

$$G_1 * G_2 \cong G_1 *_{\{e\}} G_2$$

Proof. The claim follows immediately from the following lemma.

Lemma 5.6

The free product $G_1 * G_2$ satisfies the following universal property:



□

Exercise 6. *Show that*

$$G_1 *_H G_2 \cong (G_1 * G_2) / \left\langle \varphi_1(h)\varphi_2(h)^{-1} \right\rangle$$

Remark 5.7. *We can similarly define free products $*_\alpha G_\alpha$ of arbitrarily many groups $\{G_\alpha\}_{\alpha \in I}$.*

6. Van Kampen's Theorem

***Van Kampen's Theorem* 6.1**

Let (X, p) be a connected pointed space, and suppose

$$X = \bigcup_{\alpha \in I} (U_\alpha, p)$$

where $U_\alpha \subset X$ is open and $p \in U_\alpha$. Suppose that for every $\alpha, \beta \in I$, $U_\alpha \cap U_\beta$ is path connected. Then, the map

$$\Phi : *_\alpha \pi_1(U_\alpha, p) \to \pi_1(X, p)$$

$$*_\alpha [\gamma_\alpha] \mapsto \star_\alpha [\gamma_\alpha]$$

is surjective. Moreover, if $\alpha, \beta, \delta \in I$ and $U_\alpha \cap U_\beta \cap U_\delta$ is path-connected, then

$$\pi_1(X, p) \cong *_\alpha \pi_1(U_\alpha, p) / \left\langle\left\langle i_{\alpha,\beta}(\gamma) i_{\beta,\alpha}(\gamma^{-1}) \right\rangle\right\rangle,$$

where

$$i_{\alpha,\beta} : \pi_1(U_\alpha \cap U_\beta, p) \to \pi_1(U_\alpha, p)$$

is the expected embedding, and $\langle\langle - \rangle\rangle$ is the normal closure. In particular, if $(X, p) = U_1 \cup U_2$

with $U_1 \cap U_2$ path connected, then

$$\pi_1(X,p) = \pi_1(U_1,p) \underset{\pi_1(U_1\cap U_2,p)}{*} \pi_1(U_2,p)$$

Example 6.2. *Take $X = \bigvee_{\alpha\in I}(S^1,p)$, which is the space defined by attaching I copies of S^1 at one shared point p. For each α, take $U_\alpha = S^1_\alpha \cup N_\varepsilon(p)$. Then,*

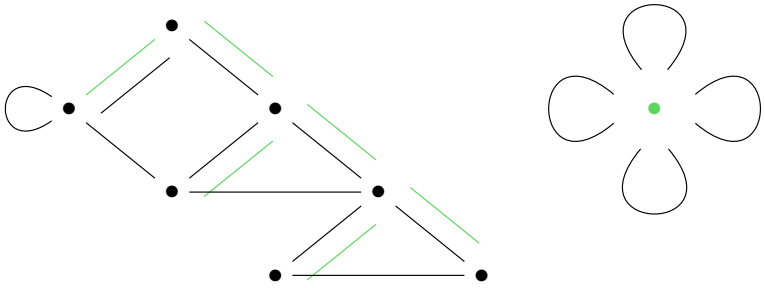
$$U_\alpha \cap U_\beta \cap U_\delta = N_\varepsilon(p)$$

for every $\alpha,\beta,\delta \in I$ is connected (and contractible). Thus, by van Kampen's theorem,

$$\pi_1(X,p) \cong *_\alpha \pi_1(S^1_\alpha,p) \cong *_\alpha \mathbb{Z}$$

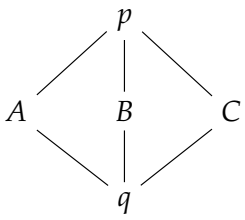
Indeed, the normal subgroup is trivial because $N_\varepsilon(p)$ is contractible.

Example 6.3. *Let $G = (V,E)$ be an undirected connected graph, viewed as a CW-complex. Let T be a spanning tree. Then, deforming the tree to a point, we get a bouquet as in the previous example:*



Remark 6.4. *Recall that we required $U_\alpha \cap U_\beta$ to be connected for surjectivity to hold. If they are not connected, then we get examples of maps which aren't surjective: for instance, taking two arclengths of S^1, their intersection has two connected components, and $\{e\} * \{e\} \not\twoheadrightarrow \mathbb{Z}$.*

The triple intersection condition is also necessary. Consider the graph:



Let $U_x = X \setminus \{x\}$. Then, $U_x \cap U_y$ is always contractible, but $U_A \cap U_B \cap U_C$ is disconnected. Notice further that $\pi_1(U_x,p) = \mathbb{Z}$, and $\pi_1(X,p) = \mathbb{Z} * \mathbb{Z}$. Then, van Kampen would say $\mathbb{Z} * \mathbb{Z} * \mathbb{Z} \twoheadrightarrow \mathbb{Z} * \mathbb{Z}$, and $\mathbb{Z} * \mathbb{Z} * \mathbb{Z} / \langle e \rangle \cong \mathbb{Z} * \mathbb{Z}$, but this is absurd.

We now prove van Kampen's theorem.

Proof of van Kampen. Let (X,p) be a pointed space, $X = \bigcup_{\alpha\in I}(U_\alpha,p)$, suppose $U_\alpha \cap U_\beta$ is path connected for all α,β, and let Φ be the map

$$\Phi : *\pi_1(U_\alpha,p) \to \pi_1(X).$$

We claim Φ is surjective. Let $\gamma : [0,1] \to X$ be a loop at p, and we want to decompose it into loops in (U_α, p). Since γ is continuous, the set

$$\{\gamma^{-1}(U_\alpha)\}_{\alpha \in I}$$

is an open cover of $[0,1]$. Thus, there exists $\delta > 0$ called the Lebesgue number, such that any open subset of $[0,1]$ with diameter $< \delta$ is contained in $\gamma^{-1}(U_\alpha)$ for some $\alpha \in I$. Assume $\delta = 1/k$ for $k \in \mathbb{N}$, and partition the interval $[0,1]$ into $2k$ sub-intervals of length $1/2k$. Thus, each sub-interval $[\frac{i-1}{2k}, \frac{i}{k}]$ is mapped via γ into U_α. Let $t_i = i/2k$ and $p_i = \gamma(t_i) \in U_{\alpha_i}$. Since $U_{\alpha_i} \cap U_{\alpha_{i+1}}$ is connected, there exists a curve μ_i from p_i to p. Let $\gamma_i = \gamma|_{[t_{i-1}, t_i]}$. Then,

$$\gamma = \gamma_{2k} * \cdots * \gamma_1 \sim \gamma_{2k} \cdots \gamma_3 \mu_2^{-1} \mu_2 \gamma_2 \mu_1^{-1} \mu_1 \gamma_1,$$

proving surjectivity.

Now, in order to prove the desired isomorphism, it suffices to show

$$\ker(\Phi) \cong \left\langle \left\langle i_{\alpha\beta}(\gamma) \cdot i_{\beta\alpha}(\gamma^{-1}) \right\rangle \right\rangle =: N.$$

Indeed, first observe that

$$\Phi(i_{\alpha\beta}(\gamma) \cdot i_{\beta\alpha}(\gamma^{-1})) = [\gamma][\gamma^{-1}] = e,$$

so clearly $N \subset \ker \Phi$.

For the other direction, we want to show that if $\gamma_1 * \cdots * \gamma_k$ is homotopic to $[e]$ as loops in X, then $\gamma_1 * \cdots * \gamma_k \in N$. This is provable, but the proof was confusing in lecture... I didn't feel like finishing it. □

Exercise 7. *Show that $\mathbb{R}^2 \setminus \{p\}$ deformation retracts to S^1, and conclude that $\mathbb{R}^2 \ncong \mathbb{R}^k$ for $k > 2$.*

Exercise 8. *Compute $\pi_1(\mathbb{R}^2 \setminus \{p_1, \ldots, p_k\})$ using van Kampen's theorem.*

Exercise 9. *How do you hang a picture using 1 rope and k nails such that removing any nail will make it fall.*

Example 6.5 (The Fundamental Group of $\mathbb{R}^3 \setminus S^1$)**.** *We can imagine the sphere of which S^1 is a cross section (S_2), and deformation retract everything outside that sphere down to it (that is, we can ignore everything fair outside our circle). We claim that $\overline{B} \setminus S^1$ deformation retracts to $S^2 \cup L$, where B is the closed ball corresponding to our sphere S_2, and L is a line passing from one end of S^1 to another.*

Then, L can be homotoped within the interior of S^2 to a closed loop touching a point of S^2. That is, our space retracts to $S^2 \vee S^1$, and by van Kampen's theorem

$$\pi_1(\mathbb{R} \setminus \{0\}) = \pi_1(S^2 \vee S^1) = \pi_1(S^1) = \mathbb{Z}.$$

Example 6.6. *Now, consider $\mathbb{R}^3$, with multiple separate circles removed. By shrinking $\mathbb{R}^3$ to the balls surrounding each circle, and using the same strategy as the previous example, we see that this space is homotopy equivalent to $S^2 \vee S^1 \vee S^2 \vee S^1 \vee \cdots \vee S^2 \vee S^1$, and so the fundamental group is $\mathbb{Z} * \mathbb{Z} * \cdots * \mathbb{Z}$.*

Example 6.7. *Now, consider $\mathbb{R}^3$, with two linked circles removed. We claim this is homotopy equivalent to $T^2 \vee S^2$, so the fundamental group is*

$$\pi_1(T^2) * \pi_1(S^2) = \mathbb{Z}^2 * \{e\} = \mathbb{Z}^2.$$

The idea is as follows: we thicken one of our circles so that it becomes a torus, and we surround our torus and circle by a sphere so that the torus touches the boundary of the sphere. Then, the second circle is "jump-roped" around the torus so that it reaches the boundary of the sphere at that same of the intersection point.

Example 6.8. *Consider the space $S^n \setminus \{p\}$, for $n \geq 0$. We claim the fundamental group is trivial. Indeed, we may pass S^{n-1} through our point, which splits S^n into an "upper-hemisphere" U and "lower-hemisphere" L. We widen U and L mildly so that they have non-trivial intersection, and $U \cap L \sim S^1$. Then,*

$$\pi_1(S^n, p) = \pi_1(U) *_{\pi_1(S^{n-1}, p)} \pi_1(L),$$

which by induction is trivial (since $\pi_1(S^1 \setminus \{p\}) = \{0\}$).

Proposition 6.9

Suppose Y is obtained from a path-connected pointed space (X, x_0) by attaching a 2-cell, i.e.

$$Y = X \sqcup \overline{D}^2 / (\varphi(x) \sim x),$$

where $X \in \partial\overline{D}^2$ and $\varphi : \partial D^2 \to X$. Let $p \in Y \cap X$ be a point such that there exists $\widetilde{p} \in D^2$ with $\varphi(\widetilde{p}) = p$. Let μ_p be a path in X between x_0 and p. Then,

$$\pi_1(Y, x_0) = \pi_1(X, x_0) / \left\langle\left\langle \mu_p^{-1} * \varphi * \mu_p \right\rangle\right\rangle.$$

Proof. Let p' be a point in the interior of D obtained by moving radially inward from $\widetilde{p}$. Let $\widetilde{\varphi}$ be a loop at p', let φ_1 be a loop in D^2 with $\widetilde{\varphi}$ in the interior, and φ_2 be a loop with $\widetilde{\varphi}$ in the exterior. Let G be the region within the loop φ_1, and P the region outside the loop φ_2. Then, p' is contained in $G \cap P$, and we may write $Y = G \cup P$. Moreover, $G \cap P$ is an annulus which is homotopy equivalent to S^1. It follows that

$$\pi_1(Y, p') = \pi_1(G, p') *_{\pi_1(S^1, p')} \pi_1(P, p').$$

Moreover, $\widetilde{\varphi}$ is homotopic to $\overline{p'p^{-1}}\varphi\overline{p'p}$, so

$$\pi_1(Y, p') \cong \pi_1(P, p') / \left\langle \overline{p'p^{-1}}\varphi\overline{p'p} \right\rangle.$$

But, note that P deformation retracts via $\mu_p\overline{p'p^{-1}}$ to give the desired result. □

Question 6.10. *What happens if we attach a k-cell for $k > 2$?*

Claim 6.11. *If $Y = X \coprod D^k / \varphi(x) \sim x$ where $x \in \partial D^k$ for $k > 2$, then $\pi_1(Y) = \pi_1(X)$. That is, "attaching k-cells with $k > 2$ does not change the fundamental group"*

Proof. Doing the same thing we did last time, by van Kampen's we have that

$$\pi_1(Y) = \pi_1(X) *_{\pi_1(S^{k-1})} \pi_1(D^k) = \pi_1(X)$$

as both $\pi_1(S^{k-1})$ and $\pi_1(D^k)$ are trivial. □

Let X be a finite path connected CW-complex.

X^0 is just points. X^1 is a connected graph which is homotopic to $\bigvee_{i=1}^{j} S_i^1$ for some j, hence $\pi_1(X^1) = \mathbb{Z}^{*j}$.

X^2 is then obtained by attaching 2-cells via

$$X^2 = X_1 \coprod \coprod_{k=1}^{j_\alpha} \overline{D}_k^2 \left(\mathrm{mod}\, x \sim \varphi_j^1(x)\right).$$

Choosing $x_0 \in X^1$, points $p_1, \cdots, p_{j_\alpha} \in \partial D_k^2$ and curves μ_j connecting x_0 to p_j we have that

$$\pi_1(X^2, x_0) = \pi_1(X^1, x_0) / \left\langle\left\langle \mu_i^{-1} \varphi_j^1 \mu_i \right\rangle\right\rangle.$$

Now $\pi_1(X) = \pi_1(X^2)$.

Remark 6.12. *This is true without assuming finiteness.*

Exercise 10 (Important!)**.** *Use van Kampen's theorem to compute*

- $\pi_1(\mathbb{T}^2)$,
- $\pi_1(\textit{Klein Bottle})$,
- $\pi_1(\mathbb{RP}^2)$,
- $\pi_1(\mathbb{RP}^k)$ *for all* k.

7. Covering Spaces

Definition **7.1**

Let X be path connected. A space $\widetilde{X}$ with a continuous map $p : \widetilde{X} \to X$ is called a **covering space** for X if for all $x \in X$ there exists an open set $U \ni x$ such that $p^{-1}(U) = \coprod_{\alpha \in I_x} U_\alpha$ for some index set I_x, where $p : U_\alpha \to U$ is a homeomorphism for all $\alpha \in I_x$.

Example 7.2. $X = S^1$, $\widetilde{X} = \mathbb{R}$, $p(\theta) = e^{i\theta}$.

Example 7.3. $X = S^1$, $\widetilde{X} = S^1$, *by* $z \mapsto z^k$ *or equivalently* $\theta \to k\theta$.

Example 7.4. $\mathbb{R}^2 \to \mathbb{R}^2/\mathbb{Z}^2 \cong \mathbb{T}^2$ *given by the natural projection.*

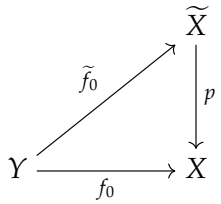
Example 7.5. $\mathbb{R}^2 \to$ *the Klein bottle given by the natural projection. "Think about it."*

Example 7.6. *We get an open cover of* $S^1 \vee S^2$ *by turning* S^1 *into a interval. Then we find another covering by "going around A at twice the speed." We of course want another covering, so we take the first covering and turn the remaining circles into intervals as well. This last covering is connected.*

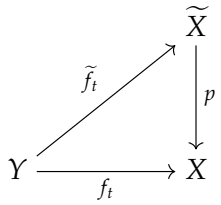
Homotopy Lifting Property **7.7**

Suppose $\widetilde{X} \to X$ is a covering map and Y is a space, $f_t : Y \times [0,1] \to X$ is a homotopy (which

inputs like $f_{.5}(y)$), and $\widetilde{f}_0 : Y \to \widetilde{X}$ is such that the following diagram commutes:



Then, there exists a unique homotopy $\widetilde{f}_t$ such that



commutes for all t.

Proof. Fix a point $y \in Y$, then $\mathrm{Im}(\overbrace{t \mapsto f_t(y)}^{\gamma_y(t)})$ is compact and for each t we can choose $U_{f_t(y)} \subset X$ as in the definition of the covering space. (a neighborhood whose preimage under p is a disjoint union of itself). Now, $\left\{\gamma_y^{-1}(U_{f_t(y)})\right\}_{t\in[0,1]}$ is a open cover of $[0,1]$, which is compact so it has a Lebesgue number. Then, we want to construct $\widetilde{\gamma}_y(t) := \widetilde{f}_t(y)$, and to do this we can trace along the interval and watch where the path "must lift".

Exercise 11. *Prove $\widetilde{f}_t$ is continuous.*

□

Proposition 7.8

Let $x_0 \in X$ be any point. Then, $p^{-1}(x_0)$ is discrete, so in particular any path lifting the constant path $\gamma \equiv x_0$ is constant.

Proof. Trivial from the definition, but also follows by the unique homotopy lifting property (Proposition 7.7). □

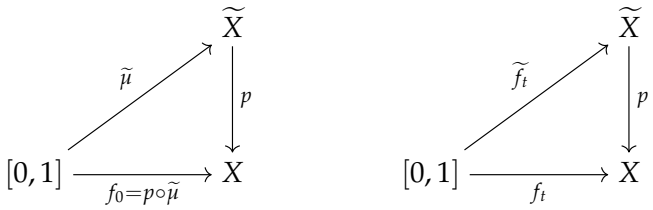
Proposition 7.9

Suppose

$$(\widetilde{X}, \widetilde{x_0}) \xrightarrow{p} (X, x_0)$$

is a covering, with X path connected, and fix $[\gamma] \in \pi_1(X, x_0)$. Then, $[\gamma] \in p_*\left(\pi_1(\widetilde{X}, \widetilde{x_0})\right)$ if and only if γ lifts to a loop at $\widetilde{x_0}$.

Proof. If γ lifts to a loop $\widetilde{\gamma}$ in $\pi_1(\widetilde{X}, \widetilde{x_0})$, then $\gamma = p \circ \widetilde{\gamma}$, hence $p_*([\widetilde{\gamma}]) = \gamma$. On the other hand, if $[\gamma] \in p_*\left(\pi_1(\widetilde{X}, \widetilde{x_0})\right)$, then γ is homotopic to $p \circ \widetilde{\mu}$ for some loop $\widetilde{\mu}$ at $\widetilde{x_0}$ in $\widetilde{X}$. Denoting by f_t this homotopy such that $f_0 = p \circ \widetilde{\mu}$ and $f_1 = \gamma$, by the homotopy lifting property, there exists a unique $\widetilde{f}_t$ lifting f_t such that $\widetilde{f}_0 = \widetilde{\gamma}$.



We claim $\widetilde{f_1}$ is a loop at $\widetilde{x_0}$ lifting γ. Indeed, as f_t is base-point preserving, $t \mapsto f_t(1)$ is a constant map. Hence, by Proposition 7.8, f_t lifts to a constant path with

$$\widetilde{f_1}(1) = \widetilde{f_0}(1) = \widetilde{\mu}(1) = \widetilde{x_0}.$$

□

Question 7.10. *When do two elements $[\gamma_1], [\gamma_2] \in \pi_1(X, x_0)$ satisfy $\widetilde{\gamma_1}(1) = \widetilde{\gamma_2}(1)$ in $p^{-1}(x_0)$, assuming $\widetilde{\gamma_1}(0) = \widetilde{\gamma_2}(0) = \widetilde{x_0}$?*

Well, knowing $\widetilde{\gamma_1}(1) = \widetilde{\gamma_2}(1)$, we may apply Proposition 7.9, hence $\widetilde{\gamma_2}^{-1} * \widetilde{\gamma_1}$ is a loop at $\widetilde{x_0}$ if and only if

$$\left[\gamma_2^{-1} * \gamma_1\right] \in p_*\left(\pi_1(X, x_0)\right) \iff \gamma_1 \in [\gamma_2]\, p_*\left(\pi_1(\widetilde{X}, \widetilde{x_0})\right),$$

i.e. when we have equivalence in the quotient space.

Corollary 7.11

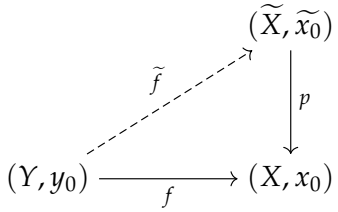
$p^{-1}(x_0)$ is in bijective correspondence with

$$\pi_1(X, x_0) / p_*\left(\pi_1(\widetilde{X}, \widetilde{x_0})\right).$$

Question 7.12. *When can f be lifted to a map*

$$\widetilde{f} : (Y, y_0) \to (\widetilde{X}, \widetilde{x_0})?$$

That is, when can me make the following diagram commute?:



Theorem 7.13

Suppose Y is path connected and locally path connected. Then, a map $f : (Y, y_0) \to (X, x_0)$ can be lifted to $\widetilde{f} : (Y, y_0) \to (\widetilde{X}, \widetilde{x_0})$ if and only if

$$f_*\left(\pi_1(Y, y_0)\right) \subset p_*\left(\pi_1(\widetilde{X}, \widetilde{x_0})\right)$$

Proof. If a lift exists, i.e. there exists $\widetilde{f}$ such that $p \circ \widetilde{f} = f$, then

$$f_* = p_* \circ \widetilde{f}_* \implies \mathrm{Img}(f_*) \subset \mathrm{Img}(p_*),$$

proving the first direction. For the other direction, fix $y \in Y$, and take a path γ connecting y_0 to y. Then, $f \circ \gamma$ is a curve in X starting at x_0, so by the homotopy lifting property there exists $\widetilde{\mu}(t)$ lifting $f(\gamma(t))$. Define $\widetilde{f}_\gamma(y) := \widetilde{\mu}(1)$. We now need to show that this $\widetilde{f}$ is independent of the choice of γ.

Let γ_1, γ_2 be two such paths from y_0 to y. Define $\mu_i(t) = f(\gamma_i(t))$, and let $\widetilde{\mu_i}$ be the lifts. We want to show $\widetilde{\mu_1}(1) = \widetilde{\mu_2}(1)$. Note that $\mu_1^{-1} \circ \mu_2$ is a loop in X at x_0, and

$$[\mu_1^{-1} * \mu_2] = f_* \left([\gamma_1^{-1} * \gamma_2] \right) \in f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\widetilde{X}, \widetilde{x_0})).$$

Thus, μ_1, μ_2, and $\mu_1^{-1} * \mu_2$ have all been lifted to $\widetilde{X}$, and these lifts are unique. It follows that $\widetilde{\mu_1}(1) = \widetilde{\mu}_2(1)$. Since we have well-definedness, we may set $\widetilde{f}(y) = \widetilde{f}_\gamma(y)$ for any γ.

Continuity is left to the reader. □

Unique Lifting Property **7.14**

Suppose Y is path connected, $f : Y \to X$ is continuous, and $\widetilde{f}, \widetilde{f}' : Y \to \widetilde{X}$ are two lifts of f. If $\widetilde{f}(y) = \widetilde{f}'(y)$ for some $y \in Y$, then $\widetilde{f} \equiv \widetilde{f}'$.

Proof. Consider the set $A := \left\{ y \,\middle|\, \widetilde{f}(y) = \widetilde{f}'(y) \right\}$, which we assume is non-empty. It suffices to show that A is clopen. Fix $y \in A$, and select $U \ni y$ as in the definition of a covering space. Let $U_\alpha \subset \widetilde{X}$ map homeomorphically to U via p, and such that $\widetilde{f}(y) = \widetilde{f}'(y) \in U_\alpha$. We claim $y \in f^{-1}(U) \subset A$. Indeed, since p is a homeomorphism from U_α to U, one sees that this is the case. □

7.1. Covering Space / Subgroup Correspondence

Proposition **7.15**

Let (X, x_0) be locally path connected, and suppose we have two coverings:

$$\begin{array}{ccc} (\widetilde{X}, \widetilde{x_0}) & & (\widetilde{X}', \widetilde{x_0}') \\ & {\scriptstyle p} \searrow \quad \swarrow {\scriptstyle p'} & \\ & (X, x_0) & \end{array}$$

Moreover, suppose

$$p_* \left(\pi_1(\widetilde{X}, \widetilde{x_0}) \right) \cong p'_* \left(\pi_1(\widetilde{X}', \widetilde{x_0}') \right).$$

Then, there exists a unique homeomorphism $f : (\widetilde{X}, \widetilde{x_0}) \to (\widetilde{X}', \widetilde{x_0}')$ such that the following diagram commutes:

$$\begin{array}{ccc} \widetilde{X} & \xrightarrow{\quad f \quad} & \widetilde{X}' \\ & {\scriptstyle p} \searrow \quad \swarrow {\scriptstyle p'} & \\ & X & \end{array}$$

Proof. Use Theorem 7.14 to lift $p : \widetilde{X} \to X$ to the cover $\widetilde{X}'$, and to lift $p' : \widetilde{X}' \to X$. These lifts are unique, and one can show they are inverses, hence the lift of p is the unique homeomorphism. □

8. Universal Covers

Definition 8.1

A **universal cover** of X is a simply connected cover of X.

Question 8.2. *Does there always exist a universal cover?*

Definition 8.3

A space X is called **semi-locally simply connected** if for every point $x \in X$, there exists a neighborhood $U \ni x$ such that the image of the map $i_* : \pi_1(U, x) \to \pi_1(X, x)$ induced by the embedding $i : U \hookrightarrow X$ is trivial.

Remark 8.4. *Simply connected implies semi-locally simply connected.*

Proposition 8.5

If X has a universal cover, then X is semi-locally simply connected.

Proof. Fix $x \in X$, and let $U \ni x$ be as in the definition of the covering space $\widetilde{X}$, which is our universal covering. Let $\tilde{x} \in \widetilde{X}$ be such that $p(\tilde{x}) = x$, and let $U_\alpha \ni \tilde{x}$ be such that $p : \widetilde{U}_\alpha \to U$ is a homeomorphism. Then, any loop γ at x in U lifts to a loop $\tilde{\gamma}$ at $\tilde{x}$ in $\widetilde{U}_\alpha$. However, since $\widetilde{U}_\alpha$ is simply connected by our assumption, we know $\tilde{\gamma}$ is homotopically trivial, and thus γ is as well. □

***Existence of the Universal Cover* 8.6**

Suppose X is path connect, locally path connected, and semi-locally simply connected. Then, X has a universal cover.

Proof. We want to construct the universal cover $\widetilde{X}$. Suppose, for now, that such a universal cover existed, and $p : \widetilde{X} \to X$ is the projection. Fix $x_0 \in X$, and $\widetilde{x_0} \in X$. Given a curve γ at x_0, we can lift it to $\widetilde{\gamma}$ starting at $\widetilde{x_0}$. Now, consider

$$\Gamma := \{\gamma : [0,1] \to X, \gamma(0) = x_0\}.$$

Then, we have an induced map $F : \Gamma \to \widetilde{X}$ given by $F(\gamma) := \widetilde{\gamma}(1)$, where $\widetilde{\gamma}$ is the correspond lift. Since $\widetilde{X}$ is path connected, F is surjective. Moreover, if $F(\gamma_1) = F(\gamma_2)$, we must have $\gamma_1(1) = \gamma_2(1)$, and $\widetilde{\gamma_2}^{-1} \circ \widetilde{\gamma_1}$ is a loop in $\widetilde{X}$ and therefore homotopically trivial, which means $\gamma_2^{-1} \circ \gamma_1$ are as well. That is, $\gamma_1 \overset{\text{rel}\{0,1\}}{\sim} \gamma_2$. Conversely, if $\gamma_1 \sim \gamma_2$, then by the homotopy lifting property, $F(\gamma_1) = F(\gamma_2)$. Thus, F induces a bijection $(\Gamma / \sim) \leftrightarrow \widetilde{X}$, where $\sim$ is homotopy relative to $\{0,1\}$. This hints that, to find our universal cover $\widetilde{X}$, we should use the collection $\Gamma \setminus \sim$.

Indeed, define

$$\widetilde{X} := \left\{ [\gamma]_{\text{rel}\{0,1\}} \,\middle|\, \gamma : [0,1] \to X, \gamma(0) = x_0 \right\},$$

and let $p : \widetilde{X} \to X$ be the projection $p([\gamma]) = \gamma(1)$, which is well-defined. Define a basis B for X

by

$$B := \{U \text{ path connected and } i_*(\pi_1(U, x_0)) \subset \pi_1(X, x_0) \text{ is trivial}\}.$$

Note that if $V \subset U \in B$ is path-connected, then $V \in B$. Thus, if $U_0 \cap U_1$ is non-trivial for $U_0, U_1 \in B$, then for any $x \in U_0 \cap U_1$, there exists a path connected $V \ni x_0$, $V \subset U_0 \cap U_1$, because X is locally path connected, and then $V \in B$. Moreover, B covers X, hence B indeed defines a basis for X.

Now, to obtain a topology for $\widetilde{X}$: for every $[\gamma] \in \widetilde{X}$ and $U \in B$ such that $\gamma(1) \in U$, take

$$U_{[\gamma]} := \{[\mu \circ \gamma] \mid \mu(0) = \gamma(1), \mathrm{Img}(\mu) \subset U\}.$$

Then, p induces a bijection $U_{[\gamma]} \to U$ for similar reasons as before. Moreover, if $\gamma_2 \in U_{[\gamma]}$, then $U_{[\gamma_2]} = U_{[\gamma]}$. Thus

$$\widetilde{B} := \left\{U_{[\gamma]} \,\middle|\, U \in B, \gamma(1) \in U\right\}$$

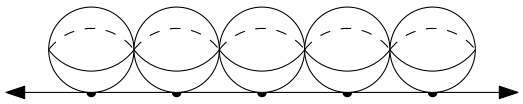
defines a basis for $\widetilde{X}$: if $U^1_{[\gamma_1]} \cap U^2_{[\gamma_2]} \neq \emptyset$, then we may take γ_3 in the intersection so that $U^1_{[\gamma_1]} = U^1_{[\gamma_3]}$ and $U^2_{[\gamma_2]} = U^2_{[\gamma_3]}$. Thus, since B is a basis for U, we may take V with $\gamma_3(1) \in V \subset U_1 \cap U_2$, and then $V_{[\gamma_3]} \subset U^1_{[\gamma_3]} \cap U^2_{[\gamma_3]} \in \widetilde{B}$, proving $\widetilde{B}$ is a basis.

One now checks that $\widetilde{X}$, with the topology induces by $\widetilde{B}$, indeed defines a covering. Finally, to prove that it's simply connected: fix $[\gamma]$ a curve on X, and define

$$[\gamma]_t(s) := \begin{cases} \gamma(s) & 0 \leq s \leq t \\ \gamma(t) & t \leq s \leq 1 \end{cases},$$

which demonstrates that $\widetilde{X}$ is path-connected. Moreover, if $\gamma \in p_*(\pi_1(X, x_0))$, and $\widetilde{\gamma}$ is the lift, then we know that $[\gamma] = [x_0]$, and so $\widetilde{\gamma}$ is homotopically trivial, proving simply connectedness. □

Example 8.7. *The universal cover of S^1 adjoined to S^2 is $\mathbb{R}$ with a copy of S^2 at every $2\pi h$, $h \in \mathbb{Z}$.*



Exercise 12. *Compute the universal cover of $S^1 \vee T^2$.*

Proposition 8.8

If $\widetilde{X}$ is a cover of X, and $x_0 \in X$, then $\pi_1(X, x_0)$ acts on $p^{-1}(x_0)$: given $\widetilde{x_0} \in p^{-1}(x_0)$, and $\gamma \in \pi_1(X, x_0)$, we may lift γ to a curve $\widetilde{\gamma}$ in $\widetilde{X}$ starting at $\widetilde{x_0}$, and so we define the action of γ on $\widetilde{x_0}$ to be $\widetilde{\gamma}(1)$.

Lemma 8.9

If $\widetilde{X}$ is the universal cover of X, then $\pi_1(X, x_0)$ acts on $\widetilde{X}$, extending the above action in a cover respecting manner: for all $g \in \pi_1(X, x_0)$, and $\widetilde{x} \in \widetilde{X}$, $p(g \cdot \widetilde{x}) = p(\widetilde{x})$.

Proof. Take $\widetilde{x_0} \in p^{-1}(x_0)$. Given $g \in \pi_1(X, x_0)$. We have the induced projections

$$\begin{array}{ccc} & & (\widetilde{X}, g\widetilde{x_0}) \\ & & \downarrow p \\ (\widetilde{X}, \widetilde{x_0}) & \xrightarrow[p]{} & (X, x_0) \end{array}$$

and so we want to find a continuous map T_g between $(\widetilde{X}, \widetilde{x_0})$ and $(\widetilde{X}, g\widetilde{x_0})$.

Given $\widetilde{x} \in \widetilde{X}$, take a curve $\widetilde{\mu}$ from $\widetilde{x_0}$ to $\widetilde{x}$, and so consider $\mu := p \circ \widetilde{\mu}$, which we lift back up to $\widetilde{X}$ starting at $g(\widetilde{x_0})$. It's left as an exercise to show that this is a group action. □

Proposition **8.10**

Let $H \leq \pi_1(X, x_0)$ be a subgroup, and denote $X_H := \widetilde{X}/H$, the space modded out by induced action of H. Then X_H is a covering space of X, with projection $p_H : X_H \to X$, and

$$p_{H*}\left(\pi_1(X_H, [\widetilde{x_0}])\right) = H$$

Proof. We have $p : (\widetilde{X}, \widetilde{x_0}) \to (X, x_0)$, the quotient map $\widetilde{p} : (\widetilde{X}, \widetilde{x_0}) \to (X_H, x_0^H)$, where $x_0^H := [\widetilde{x_0}]$, and we have $p_H : (X_H, x_0^H) \to (X, x_0)$. We claim p_H is a covering map.

Let $x \in X$, and let $U \ni x$ be path connected such that $p^{-1}(U) = \coprod U_\alpha$, with $p : U_\alpha \to U$ a homeomorphism.

Claim 8.11. *For $g \in \pi_1(X, x_0)$, if $g(U_\alpha) \cap U_\beta \neq \emptyset$, then $g(U_\alpha) = U_\beta$.*

Indeed, let $\widetilde{z_0} \in U_\alpha$ be such that $g\widetilde{z_0} \in U_\beta$. Since U_α and U are path connected, we may take $z_0 := p(\widetilde{z_0})$, and then for all $\widetilde{z} \in U_\alpha$, we take a path $\widetilde{\gamma}$ in U_α connection $\widetilde{z_0}$ and $\widetilde{z}$. Take $\gamma := p \circ \widetilde{\gamma}$, and lift γ to a path starting at $g\widetilde{z_0}$. We claim that $g \cdot \widetilde{\gamma}$ is such a lift. Thus, $g\widetilde{z} \in U_\beta$. Showing the reverse inclusion finishes the claim... □

Definition **8.12**

Let $\widetilde{X} \xrightarrow{p} X$ be a covering map, and let $T : \widetilde{X} \to \widetilde{X}$ be continuous such that $p \circ T = p$. Then, T is called a **deck transformation**.

Definition **8.13**

Let $\widetilde{X}$ be a topological space, and G act on $\widetilde{X}$ on the right via continuous maps. The action of G is called **properly discontinuous** if for all $\widetilde{x} \in \widetilde{X}$, there exists $\widetilde{U} \ni \widetilde{x}$ such that for all $g \in G$, $g(\widetilde{U}) \cap \widetilde{U} \neq \emptyset \implies g = e$.

Theorem **8.14**

Let $\left(\widetilde{X}, \widetilde{x_0}\right)$ be a simply connected, locally path connected, path connected space. Let G act on $\widetilde{X}$ properly discontinuously, and set $X := \widetilde{X}/G$ with the natural projection $p : \widetilde{X} \to X$. Then,

$\widetilde{X}$ is the universal cover of x, and setting $x_0 = p(\widetilde{x}_0)$ gives

$$\pi_1(X, x_0) \simeq G$$

Proof. Take $x \in X$, $\widetilde{x} \in p^{-1}(x)$, and let $\widetilde{U}$ be a neighborhood of $\widetilde{x}$ as in the definition, and $U = p(\widetilde{U})$. Then, $p : \widetilde{U} \to U$ is a homeomorphism, amd $p^{-1}(U) = \bigsqcup_{g\in G} g\widetilde{U}$. Thus, $\widetilde{X}$ is the universal cover, and both $\pi_1(X, x_0)$ and G act on $\widetilde{X}$ by deck transformations. For all $\widetilde{x}_0' \in p^{-1}(x_0)$, there exists a unique $g \in G$ such that $g(\widetilde{x}_0) = \widetilde{x}_0'$, and there exists a unique $\gamma \in \pi_1(X, x_0)$ such that $F_\gamma(\widetilde{x}_0) = \widetilde{x}_0' = F\gamma = g$. □

Example 8.15. *Recall that $\mathbb{RP}^n = S^n / \{x \sim (-x)\}$. Observe that $\mathbb{Z}/2\mathbb{Z}$ acts on S^n by $x \mapsto -x$, and S^n is simply connected for $n \geq 2$. The quotient of this action on S^n is exactly $\mathbb{RP}^n$, hence by Theorem 8.14, $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$.*

Exercise 13. *Use the universal cover to compute the fundamental group of the Klein bottle.*

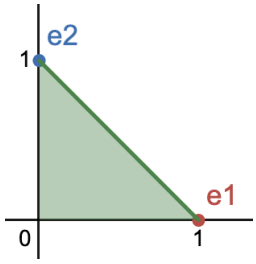
9. Simplicial Homology

Definition 9.1

Let $v_0, \ldots, v_n \in \mathbb{R}^n$ be ordered points such that no points $v_{i_1}, \ldots, v_{i_k}$ lie in any $k-1$ dimensional affine space. The n-**simplex** $[v_0, \ldots, v_n]$ is the convex hull of $v_0, \ldots, v_n$. That is,

$$[v_0, \ldots, v_n] = \left\{ \sum t_i v_i \,\middle|\, t_i \geq 0, \sum t_i = 1 \right\}$$

Example 9.2. *The **standard n-simplex** $\Delta^n \subset \mathbb{R}^{n+1}$ is the simplex $[e_1, \ldots, e_{n+1}]$. For instance, the standard 1-simplex is:*



Definition 9.3

Given an n-simplex $[v_0, \ldots, v_n]$, the i-th **face** is the $(n-1)$-simplex $[v_0, \ldots, v_{i-1}, v_{i+1}, \ldots, v_n]$. The **boundary** $\partial[v_0, \ldots, v_n]$ of a simplex is the union of its faces. The **interior**, is union of the simplex minus the boundary.

Definition 9.4

A topological space X is called a Δ-**complex** if there exists a collection of continuous maps $\left\{\sigma_\alpha : \Delta^{n(\alpha)} \to X\right\}_{\alpha\in I}$ with the following properties:

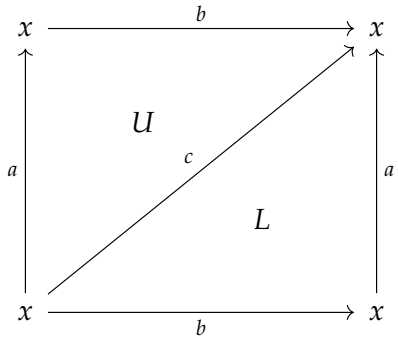
(i) For every $\alpha \in I$, the restriction $\sigma_\alpha|_{\mathring{\Delta}^{n(\alpha)}}$ is injective, and for all $x \in X$, there exists a unique α such that $x \in \mathrm{Img}(\sigma_\alpha|_{\mathring{\Delta}^{n(\alpha)}})$.

(ii) $U \subset X$ is open if and only if $\sigma_\alpha^{-1}(U)$ is open for all $\alpha \in I$

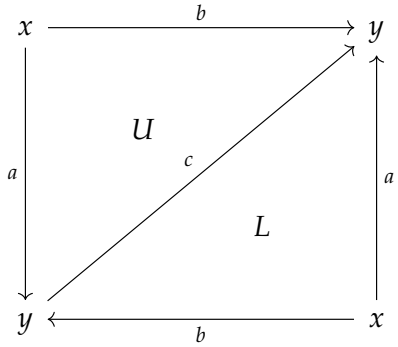
(iii) For all $\alpha \in I$, and every $0 \leq i \leq n(\alpha)$, the restriction of σ_α to the i-th face is in the collection of maps. That is, there exists $\beta \in I$ such that $\sigma_\beta = \sigma_\alpha|_{[e_0,\ldots,\widehat{e_i},\ldots,e_{n(\alpha)+1}]}$. This is the case if and only if, denoting by $\iota_i : \Delta^{n(\alpha)-1} \to \Delta^{n(\alpha)}$ the map sending $(x_0,\ldots,x_{n(\alpha)-1}) \mapsto (x_0,\ldots,x_{i-1},0,x_{i+1},\ldots,x_{n(\alpha)-1})$, we have $\sigma_\beta = \sigma_\alpha \circ \iota_i$.

Example 9.5 (Δ-complex structure on S^1)**.** *Simple...*

Example 9.6 (Δ-complex structure of $\mathbb{T}^2$)**.** *We have a single 0-complex x (the point representing the four corners of the rectangle), two 1-simplices a, b representing the attached sides of the rectangle. Note that the complex Δ^1 is a triangle, but we're working with a rectangle. We fix this by taking another line c attaching opposite corners...*



Example 9.7 (Δ-complex structure of $\mathbb{RP}^2$)**.** *Similar to $\mathbb{T}^2$...*



Exercise 14. *Find a Δ-complex structure on S^2 and the Klein bottle.*

Definition **9.8**

Let X be a Δ-complex, with simplices $\{\sigma_\alpha : \Delta^{n(\alpha)} \to X\}_{\alpha\in I}$, let $I(n) = \{\alpha \mid n(\alpha) = n\}$, and $I = \bigcup I(n)$. The **group of simplicial n-chains** $\Delta_n(X)$ is the free abelian group on $\{\sigma_\alpha\}_{\alpha\in I(n)}$. That is,

$$\Delta_n(x) = \left\{ \sum_{\alpha\in I(n)} m_\alpha \sigma_\alpha \;\middle|\; m_\alpha \in \mathbb{Z}, m_\alpha = 0 \text{ for all but finitely many } \alpha \right\}.$$

Such an expression $\sum m_\alpha \sigma(\alpha)$ is called a **simplicial chain**.

Definition 9.9

The nth **boundary map** $\partial_n : \Delta_n(X) \to \Delta_{n-1}(X)$ is the homomorphism defined on the basis by

$$\partial_n(\sigma_\alpha) = \sum_{i=0}^{n} (-1)^i \sigma_\alpha|_{[e_0,\ldots,\widehat{e_i},\ldots,e_n]}$$

Proposition 9.10

$\partial_{n-1} \circ \partial_n \equiv 0$.

Proof. We break the sum up corresponding to when the second removed basis element comes before or after the first:

$$\begin{aligned}\partial_{n-1} \circ \partial_n(\sigma_\alpha) &= \partial_{n-1}\left(\sum_{i=0}^{n} (-1)^i \sigma_\alpha|_{[e_0,\ldots,\widehat{e_i},\ldots,e_n]} = \right) \\ &= \sum_{j=0}^{i-1} (-1)^{i+j} \sigma_\alpha|_{[e_0,\ldots,\widehat{e_i},\ldots,\widehat{e_j},\ldots,e_n]} + \sum_{j=i+1}^{n} (-1)^{i+j+1} \sigma_\alpha|_{[e_0,\ldots,\widehat{e_j},\ldots,\widehat{e_i},\ldots,e_n]}.\end{aligned}$$

Now, for every $k < \ell \in \{0,\ldots,n\}$, the $(n-2)$-simplex

$$\sigma_\alpha|_{[e_0,\ldots,\widehat{e_k},\ldots,\widehat{e_\ell},\ldots,e_n]}$$

appears twice, for $(k,\ell) = (i,j)$ and $(k,\ell) = (j,i)$, but they appear with opposite sign. Therefore, they cancel, and the resulting sum is zero. □

Definition 9.11

A **chain complex** is a sequence of abelian groups $\{C_n\}_{n=0}^{\infty}$ with homomorphisms $\partial_n : C_n \to C_{n-1}$ such that $\partial_{n-1} \circ \partial_n \equiv 0$.

The elements of $\ker(\partial_n) \subset C_n$ are called n**-cycles**, and the elements of $\operatorname{Img}(\partial_{n+1}) \subset C_n$ are called n**-boundaries**.

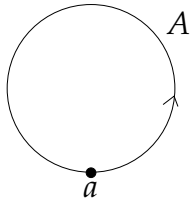
Note that the condition $\partial_n \circ \partial_{n+1} \equiv 0$ means that every n-boundary is an n-cycle, since $\operatorname{Img}(\partial_{n+1}) \subset \ker(\partial_n)$. The n-th **homology group** is the quotient $H_n(C) := \ker(\partial_n)/\operatorname{Img}(\partial_{n+1})$.

Definition 9.12

By Proposition 9.10, the simplicial n-chains $\{\Delta_n(X)\}_{n=0}^{\infty}$ along with the boundary maps $\partial_n : \Delta_n(X) \to \Delta_{n-1}(X)$ form a chain complex. The n-th **simplicial homology group** of X is

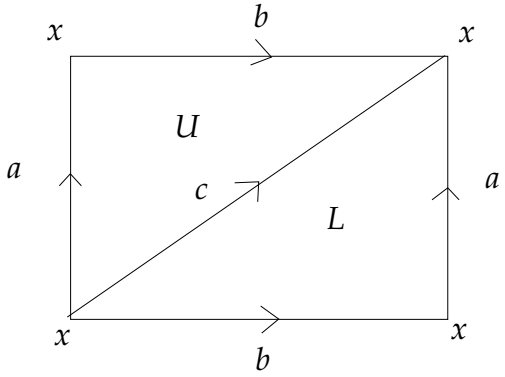
$$H_n^{\Delta}(X) := \ker(\Delta_{n+1}(X))/\operatorname{Img}(\Delta_n(X))$$

Example 9.13. *Let $X = S^1$, with the simplicial structure*



Observe that $\Delta^1(X) \simeq \mathbb{Z}$ because we have a single 1-simplex, and $\partial_1 \equiv 0$, so $H_1^\Delta(S^1) = \mathbb{Z}$. For $n \geq 2$, the homology group is $H_n^\Delta(S^1) \simeq 0$ since there are no n-simplices. It then follows that $H_0^\Delta(S^1) \simeq \mathbb{Z}$ because the homology groups form a chain complex.

Example 9.14. *Let $X = \mathbb{T}^2$.*



Similar to before, we have $H_k^\Delta(\mathbb{T}^2) = 0$ for $k \geq 3$ because that's the highest level of simplex we have.

Now, we observe that the two 2-simplices U and L have the same image under the boundary map:

$$\partial_2 U = b - c + a,$$

$$\partial_2 L = a - c + b,$$

Since U and L generate $\Delta_2(X)$, it follows that $\ker(\partial_2) = \langle U - L \rangle \simeq \mathbb{Z}$, i.e. the free abelian group on 1 generator. Moreover, there are no 3-simplices so $\operatorname{Img}(\partial_3) = 0$, and we deduce that

$$H_2^\Delta(\mathbb{T}^2) = \ker(\partial_2)/\operatorname{Img}(\partial_3) = \mathbb{Z}.$$

Next, we have three generators of $\Delta_1(X)$, namely a, b, c. Observe that they all satisfy $\partial_1 a = \partial_1 b = \partial_1 c = x - x = 0$, i.e. all three generators are in $\ker \partial_1$, so $\ker(\partial_1) \simeq \mathbb{Z}^3$. Moreover, we computed that $\partial_2 U = \partial_2 L$, so $\operatorname{Img}(\partial_2) = \mathbb{Z}$, thus

$$H_1^\Delta(\mathbb{T}^2) = \ker(\partial_1)/\operatorname{Img}(\partial_2) \simeq \mathbb{Z}^3/\mathbb{Z} \simeq \mathbb{Z}^2.$$

Finally, x is the only 0-simplex, and is trivially in the kernel of ∂_0. Moreover, we saw everything was in the kernel of ∂_1, so $\operatorname{Img}(\partial_1) = 0$, and we conclude that

$$H_0^\Delta(\mathbb{T}^2) = \ker(\partial_0)/\operatorname{Img}(\partial_1) \simeq \mathbb{Z}$$

Exercise 15. *Compute $H_n^\Delta(X)$ for the Klein bottle, $\mathbb{RP}^2$, and S^2.*

Remark 9.15. *There are generally a few problems with simplicial homology. First, it's not immediately clear that the simplicial homology groups are independent of the choice of Δ-structure. While this is true, it's tedious to show. Furthermore, doing computations with a Δ-structure can be annoying, especially as we're taking more and more complicated spaces. We therefore look for a better solution.*

10. Singular Homology

Definition 10.1

Let X be a topological space. The group of **singular n-chains** in X is the free abelian groups on

$$\{\sigma : \Delta^n \to X \mid \sigma \text{ continuous}\}.$$